ECON 2050 – Mathematical Economics

Optimization II: Linear Programming

Linear programming

In this lecture we look at a special class of optimization problems.

The problem of optimizing a *linear function* subject to *linear inequality and equality constraints* is called **linear programming** (LP).

A (running) example of linear programming problem ("linear program" for short):

$$\begin{array}{ll} \max & x_1 + 2x_2 \\ \text{s.t.} & x_1 + 3x_2 \leq 4, \\ & x_1 + x_2 \leq 2, \\ & x_1, x_2 \geq 0. \end{array}$$

In other words, in LP the objective function is linear, and the constraint set is delineated by a set of linear inequalities and equalities.

Example: (For simplicity, the numbers are unrealistic.) A bakery makes both bread and cakes.

- From selling a box of bread, the bakery makes profits of \$1, and the profits from selling a box of cakes is \$2.
- Each box of bread requires 1 unit of labor, and each box of cakes requires 3 units of labor; there is 4 units of labor available in total.
- The production capacity of the bakery limits it to produce no more than 2 boxes of bread and cakes.

Question: how many boxes of bread and cakes should be produced to maximize the profits of the bakery?

Denote the numbers of boxes of bread and cakes by $x_1 \geq 0$ and $x_2 \geq 0$, respectively.

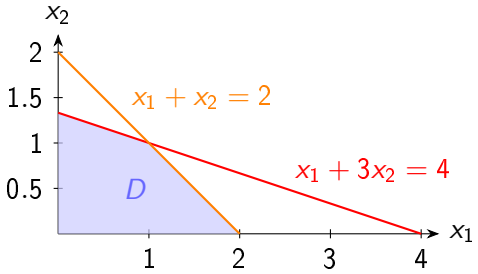
- Profits: $x_1 + 2x_2$
- Labor constraint: $x_1 + 3x_2 \leq 4$
- Production capacity constraint: $x_1 + x_2 \leq 2$

Therefore, the bakery's profit maximization problem is exactly the linear program on the previous slide.

The linear program can also be written as $\max_{(x_1,x_2)\in D} x_1 + 2x_2$, where

$$D = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 + 3x_2 \leq 4, x_1 + x_2 \leq 2, x_1 \geq 0, x_2 \geq 0\}.$$

We call a point in the constraint set **feasible** for the linear program.



Observe that D is a convex set; more generally, if all constraints are linear inequalities and equalities, the constraint set is convex.

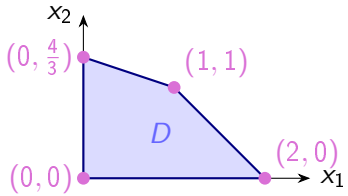
- In other words, the constraint set of any linear program is convex.

Extreme points

Definition

Let $S \subseteq \mathbb{R}^n$ be a convex set. An **extreme point** of S is a point that cannot be expressed as a convex combination of any other points in S.

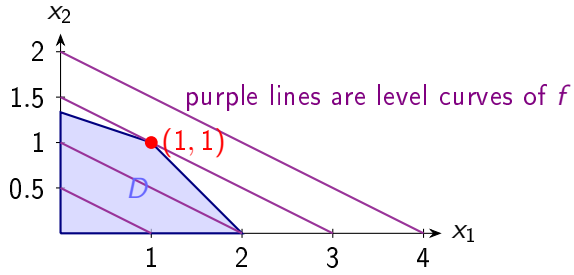
Example: The extreme points of $D = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 + 3x_2 \leq 4, x_1 + x_2 \leq 2, x_1 \geq 0, x_2 \geq 0\}$ are $(0,0)$, $(0, \frac{4}{3})$, $(2,0)$, and $(1,1)$.



In linear programming, the constraint set is a polyhedron (i.e., a generalized polygon), and its extreme points are the "corners" or "vertices."

Define $f(x_1, x_2) = x_1 + 2x_2$. Since $\nabla f(x_1, x_2) = (1, 2) \neq (0, 0)$ for every $(x_1, x_2) \in \mathbb{R}^2$, there is no critical points in the interior of D.

- This fact is more generally true for any nontrivial linear program.
- Therefore, to search for solutions/maxima, it suffices to look at boundary points.
- Graphical analysis reveals that the unique solution/maximum is extreme point $(1, 1)$.



In fact, for any linear program that has a solution, an extreme point of the constraint set must be a solution.

General setup and terminology

Every linear programming problem can be written in the following standard form:

$$\begin{aligned} \max \quad & \boldsymbol{c} \cdot \boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b} \\ & \boldsymbol{x} \geq 0 \end{aligned}$$

Or more compactly, $\max\{\boldsymbol{c} \cdot \boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}, \boldsymbol{x} \geq 0\}$.

- If x_j doesn't have to be non-negative, substitute $x_j = x_j^+ - x_j^-$, where $x_j^+, x_j^- \geq 0$.
- If the objective is $\min \boldsymbol{c} \cdot \boldsymbol{x}$ then replace it with $\max -\boldsymbol{c} \cdot \boldsymbol{x}$.
- If the i-th constraint is $\sum_{j=1}^n a_{ij}x_j \geq b_i$, replace it by $-\sum_{j=1}^n a_{ij}x_j \leq -b_i$.
- If the i-th constraint is $\sum_{j=1}^n a_{ij}x_j = b_i$, replace it by two inequality constraints: $\sum_{j=1}^n a_{ij}x_j \leq b_i$ and $-\sum_{j=1}^n a_{ij}x_j \leq -b_i$.

Suppose $\tilde{\boldsymbol{x}}$ is in the constraint set. Then if a constraint holds with equality, say that this constraint **binds** at $\tilde{\boldsymbol{x}}$; if instead the constraint holds with strict inequality, we say it is **slack** at $\tilde{\boldsymbol{x}}$.

Example: Let us revisit the example at the beginning:

$$\begin{array}{ll} \max & x_1 + 2x_2 \\ \text{s.t.} & x_1 + 3x_2 \leq 4, \\ & x_1 + x_2 \leq 2, \\ & x_1, x_2 \geq 0. \end{array}$$

In this example, $\boldsymbol{c} = (1, 2)$, $\boldsymbol{A} = \begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix}$, and $\boldsymbol{b} = (4, 2)$.

Recall that a linear function is continuous, and therefore if the constraint set

$$D = \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}, \boldsymbol{x} \geq 0\}$$

is compact, the linear program has a solution.

Fundamental theorem of linear programming

First, we formalize the observation suggested by the example: A maximum is an extreme point.

Theorem (Fundamental Theorem of Linear Programming)

Suppose the linear program $\max\{\boldsymbol{c} \cdot \boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}, \boldsymbol{x} \geq 0\}$ *has a solution. Then an extreme point of the constraint set is a solution.*

Example: Revisiting the example from the beginning:

- Observe that the constraint set is compact, and therefore a solution exists.
- By the fundamental theorem of linear programming, it suffices to search within the set of extreme points of the constraint set D.
- The extreme points are $(0,0)$, $(0,\frac{4}{3})$, $(2,0)$, and $(1,1)$, and

$$f(0,0) = 0, \quad f\left(0,\frac{4}{3}\right) = \frac{8}{3}, \quad f(2,0) = 2, \quad \text{and} \quad f(1,1) = 3.$$

- Thus, $(1,1)$ is a solution.

Going back to our bakery example,

$$\begin{array}{ll} \max & x_1 + 2x_2 \\ \text{s.t.} & x_1 + 3x_2 \leq 4, \\ & x_1 + x_2 \leq 2, \\ & x_1, x_2 \geq 0. \end{array}$$

In an attempt to solve this problem, we can produce upper bounds on its optimal value V:

- Since $x_1 + 2x_2 \leq x_1 + 3x_2 \leq 4$, we know $V \leq 4$.
- Since $x_1 + 2x_2 \leq (2/3)(x_1 + 3x_2) + (1/3)(x_1 + x_2) \leq (2/3)4 + (1/3)2 = 10/3$, $V \leq 10/3$.
- Since $x_1 + 2x_2 \leq (1/2)(x_1 + 3x_2) + (1/2)(x_1 + x_2) \leq (1/2)4 + (1/2)2 = 3$, we know $V \leq 3$.

In each of these cases we take a non-negative linear combination of the constraints, looking for better and better upper bounds on the maximum possible value of the objective function $x_1 + 2x_2$.

We can formalize this process by letting y_1 and y_2 be the coefficients of our linear combination. We must have

$$\begin{aligned} y_1 + y_2 &\geq 1, \\ 3y_1 + y_2 &\geq 2, \\ y_1, y_2 &\geq 0, \end{aligned}$$

and we seek to minimize $4y_1 + 2y_2$.

We obtained another linear program!

- We refer to this new linear program as the **dual** and the original one as the **primal**.

Recall that we designed the dual to serve as a method of constructing an upper bound on the optimal value of the primal.

- If (y_1, y_2) is feasible for the dual, and (x_1, x_2) is feasible for the primal, then by construction $x_1 + 2x_2 \leq 4y_1 + 2y_2$.
- If we can find feasible (x_1^*, x_2^*) and (y_1^*, y_2^*) that make these equal, we know we have found the optimal values of both programs. In fact, $(x_1^*, x_2^*) = (1, 1)$ and $(y_1^*, y_2^*) = (1/2, 1/2)$.

Duality

More generally, consider the standard form of a linear programming problem

$$\max \boldsymbol{c} \cdot \boldsymbol{x} \qquad \text{s.t. } \boldsymbol{Ax} \leq \boldsymbol{b},\ \boldsymbol{x} \geq 0. \tag{P}$$

Taking a linear combination of the constraints with weights $\boldsymbol{y} = (y_1, \ldots, y_m)$ can be written as $\boldsymbol{yA}$; since we need to find an upper bound, it must be that $\boldsymbol{yA} \geq \boldsymbol{c}$.

- Since $\boldsymbol{x} \geq 0$, we have $\boldsymbol{yAx} \geq \boldsymbol{c} \cdot \boldsymbol{x}$.
- Then because $\boldsymbol{Ax} \leq \boldsymbol{b}$, $\boldsymbol{c} \cdot \boldsymbol{x} \leq \boldsymbol{yAx} \leq \boldsymbol{y} \cdot \boldsymbol{b}$. Thus, $\boldsymbol{y} \cdot \boldsymbol{b}$ is an upper bound on objective function value.

Therefore, the problem of finding the smallest such upper bound is

$$\min \boldsymbol{y} \cdot \boldsymbol{b} \qquad \text{s.t. } \boldsymbol{yA} \geq \boldsymbol{c},\ \boldsymbol{y} \geq 0. \tag{D}$$

We call problem (D) the **dual** and problem (P) the **primal**.

- Denote the optimal values of (P) and (D) by V and W, respectively.
- Since $\boldsymbol{c} \cdot \boldsymbol{x} \leq \boldsymbol{y} \cdot \boldsymbol{b}$, we have $V \leq W$; this is known as **weak duality**.

Duality theorem

Theorem (Strong duality)

If either the primal or the dual has a solution, then $V = W$.

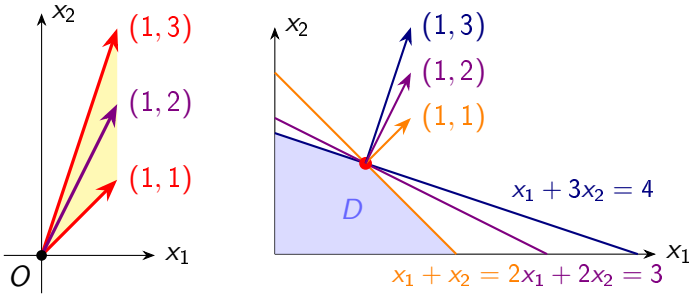
Some geometric intuition based on the bakery example:

- Recall that the bakery's problem can be written as $\max \boldsymbol{c} \cdot \boldsymbol{x}$ s.t. $\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$, $\boldsymbol{x} \geq 0$ by letting

$$\boldsymbol{c} = \begin{pmatrix} 1 & 2 \end{pmatrix}, \quad \boldsymbol{A} = \begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix}, \quad \text{and} \quad \boldsymbol{b} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}.$$

- Denote the first and the second row of $\boldsymbol{A}$ by $\boldsymbol{a}_1$ and $\boldsymbol{a}_2$, respectively.
- At the solution $\boldsymbol{x}^* = (x_1^*, x_2^*) = (1, 1)$, both "resource constraints" (labor and capacity) bind.
- Also, $\boldsymbol{a}_1$ and $\boldsymbol{a}_2$ are the "directions of increase" of the two resource constraints, respectively; and $\boldsymbol{c}$ is the direction of increase of the level curve of the objective function

Geometric intuition continued:



- The discussion in the previous slide implies that $\boldsymbol{c}$ is in the cone generated by $\{\boldsymbol{a}_1, \boldsymbol{a}_2\}$.
- Therefore, there exists $\boldsymbol{y}^* = (y_1^*, y_2^*) \geq 0$ such that $\boldsymbol{c} = y_1^* \boldsymbol{a}_1 + y_2^* \boldsymbol{a}_2$.
- Since the two resource constraints bind at $\boldsymbol{x}^*$, which can be written as $\boldsymbol{a}_1 \cdot \boldsymbol{x}^* = b_1$ and $\boldsymbol{a}_2 \cdot \boldsymbol{x}^* = b_2$, where $\boldsymbol{b} = (b_1, b_2)$.
- But then $\boldsymbol{c} \cdot \boldsymbol{x}^* = (y_1^* \boldsymbol{a}_1 + y_2^* \boldsymbol{a}_2) \cdot \boldsymbol{x}^* = y_1^* \boldsymbol{a}_1 \boldsymbol{x}^* + y_2^* \boldsymbol{a}_2 \boldsymbol{x}^* = y_1^* b_1 + y_2^* b_2 = \boldsymbol{y}^* \cdot \boldsymbol{b}$, implying $V = W$.

A proof of the duality theorem

Suppose that the primal has a solution; the argument starting from the dual having a solution is analogous. By weak duality, $V \leq W$; it suffices to show that $V \geq W$.

- If there exists a vector $\boldsymbol{x} \geq 0$ such that $\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$ and $\boldsymbol{c} \cdot \boldsymbol{x} \geq W$, then we have $V \geq W$.
- In other words, it suffices to show that the system of inequalities $\boldsymbol{c} \cdot \boldsymbol{x} \geq W$, $\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$, and $\boldsymbol{x} \geq 0$ has a solution. Also, $\boldsymbol{c} \cdot \boldsymbol{x} \geq W$ is equivalent to $-\boldsymbol{c} \cdot \boldsymbol{x} \leq -W$.
- Suppose, toward contrary, that the system $-\boldsymbol{c} \cdot \boldsymbol{x} \leq -W$, $\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$, and $\boldsymbol{x} \geq 0$ has no solution; equivalently, the system

$$\begin{pmatrix} \boldsymbol{A} \\ -\boldsymbol{c} \end{pmatrix} \boldsymbol{x} \leq \begin{pmatrix} \boldsymbol{b} \\ -W \end{pmatrix}$$

  has no non-negative solution.
- Then its Farkas alternative

$$\begin{pmatrix} \boldsymbol{y} & \lambda \end{pmatrix} \begin{pmatrix} \boldsymbol{A} \\ -\boldsymbol{c} \end{pmatrix} \geq 0, \quad \begin{pmatrix} \boldsymbol{y} & \lambda \end{pmatrix} \begin{pmatrix} \boldsymbol{b} \\ -W \end{pmatrix} < 0, \quad \text{and} \quad \begin{pmatrix} \boldsymbol{y} & \lambda \end{pmatrix} \geq 0.$$

  must have a solution.

- The Farkas alternative can be written as

$$\boldsymbol{yA} - \lambda \boldsymbol{c} \geq 0, \quad \boldsymbol{y} \cdot \boldsymbol{b} - \lambda W < 0, \quad \boldsymbol{y} \geq 0, \quad \text{and} \quad \lambda \geq 0;$$

  denote a solution by $\begin{pmatrix} \boldsymbol{y}^* & \lambda^* \end{pmatrix}$.
- Observe that $\lambda^* > 0$: if instead $\lambda^* = 0$, the system above reduces to

$$\boldsymbol{yA} \geq 0, \quad \boldsymbol{y} \cdot \boldsymbol{b} < 0, \quad \text{and} \quad \boldsymbol{y} \geq 0,$$

  which is the Farkas alternative of the system $\boldsymbol{Ax} \geq \boldsymbol{b}$ and $\boldsymbol{x} \geq 0$.
- Since it has a solution $\boldsymbol{y}^*$, the Farkas lemma implies that the system $\boldsymbol{Ax} \geq \boldsymbol{b}$ and $\boldsymbol{x} \geq 0$ has no solution, violating the assumption that the primal has a solution. Thus, $\lambda^* > 0$.
- Let $\hat{\boldsymbol{y}} \equiv \boldsymbol{y}^*/\lambda^*$; this is well-defined because $\lambda^* > 0$. The system at the top implies

$$\hat{\boldsymbol{y}} \cdot \boldsymbol{A} \geq \boldsymbol{c}, \quad \hat{\boldsymbol{y}} \geq 0, \quad \text{and} \quad \hat{\boldsymbol{y}} \cdot \boldsymbol{b} < W,$$

  showing that $\hat{\boldsymbol{y}}$ is feasible for the dual and the value of the dual objective function at $\hat{\boldsymbol{y}}$ is strictly below W, contradicting the assumption that W is the optimal value of the dual.

Interpreting the dual

A benefit of studying the dual is that it can help with solving the primal problem.

- For example, a problem with many variables and a few constraints can be reduced to another with a few variables and many constraints, which is usually much simpler.
- The "Lagrangian approach" of constrained optimization you learned in ECON1050 and/or ECON2010 is a generalization of duality: the "Lagrange conditions" stem from the dual of the optimization problem we study.

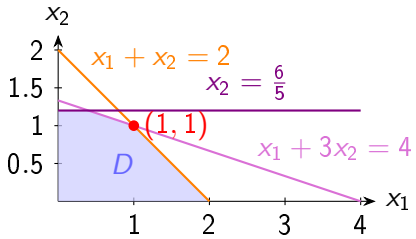
An arguably more powerful aspect of duality theory is that; however, the dual reveals valuable insights that are not immediately apparent in the primal.

- For studying economics, finance, and business, this aspect is usually more important than the computational benefits mentioned above.
- In what follows we illustrate this by going back once again to the bakery example.

Consider the example of a bakery on slide 3, with an extra twist that since cakes need to be refrigerated, there is an extra "storage constraint" that the at most 6/5 boxes of cakes can be produced in a day. Thus, the linear program becomes

$$\begin{aligned} \max \quad & x_1 + 2x_2 \\ \text{s.t.} \quad & x_1 + 3x_2 \leq 4, \\ & x_1 + x_2 \leq 2, \\ & x_2 \leq \tfrac{6}{5}, \\ & x_1, x_2 \geq 0. \end{aligned}$$

It can be verified that the unique optimal solution is still $(x_1^*, x_2^*) = (1, 1)$.



Now suppose an entrant wishes to buy the bakery from the current owner. She offers a per unit price for each resource that the owner possesses.

- Suppose the entrant intends to offer prices y_1, y_2, and y_3 for each unit of labor, production capacity, and storage space.
- Because a box of bread requires a unit of labor, a unit of production capacity, and no (fridge) storage space, the owner will get $y_1 + y_2$ for all the resources needed to produce a box of bread.
- There was originally \$1 profit for each box of bread, so for this to be a profitable transaction, the owner requires $y_1 + y_2 \geq 1$.
- Similarly, because a box of cakes requires 3 units of labor, a unit of production capacity, and a unit of (fridge) storage space, the owner will get $3y_1 + y_2 + y_3$ for all the resources needed to produce a box of cakes.
- There were originally \$2 profits for each box of cakes, so for this to be a profitable transaction, the owner requires $3y_1 + y_2 + y_3 \geq 2$.

Now the entrant wants to buy the owner's resources as inexpensively as possible while having the owner accept the offer.

- The total cost of 4 units of labor, 2 units of production capacity, and 6/5 units of storage space, is $4y_1 + 2y_2 + \frac{6}{5}y_3$.
- Therefore, the entrant's problem is

$$\begin{aligned} \min \quad & 4y_1 + 2y_2 + \tfrac{6}{5}y_3 \\ \text{s.t.} \quad & y_1 + y_2 \geq 1, \\ & 3y_1 + y_2 + y_3 \geq 2, \\ & y_1, y_2, y_3 \geq 0. \end{aligned}$$

- It is not difficult to verify that the entrant's problem is the dual of the owner's revenue maximization problem.

A consequence of the duality theorem is that the minimum total price the entrant must pay for the owner to accept the offer is exactly the maximum revenue that the owner can obtain from running the bakery.

This is not too surprising:

- For the owner to part with the bakery, he must receive at least as much money as it makes by producing and selling bread and cakes.
- The entrant, on the other hand, should pay no more than the revenue it could generate by acquiring the bakery.

The story above allows us to interpret the dual variables as "prices" for each of the resources, or beyond the acquisition setting, the "imputed cost" of using an extra unit of each of the resources.

Another way to understand the dual is to ask: would additional resource be valuable, and if so just how much?

- The solution of the dual is $(y_1^*, y_2^*, y_3^*) = (\frac{1}{2}, \frac{1}{2}, 0)$.
- Note that the constraint on the fridge storage space $x_2 \leq \frac{6}{5}$ is slack at the primal solution $(x_1^*, x_2^*) = (1, 1)$.
- In other words, as long as the fridge storage space is no less than 1, the solution $(x_1^*, x_2^*) = (1, 1)$ is still revenue-maximizing.
- Therefore, for *slight* changes in the fridge storage space, the value of y_3^* gives us the change in optimal objective function value, namely zero.
- Let's now turn to a binding constraint, say the labor constraint, and suppose the units of available labor increases by very little, say 0.1 (so 4 becomes 4.1).
- One can verify that after this change, the primal solution is still at the intersection of the (new) "labor boundary" $x_1 + 3x_2 = 4.1$ and "production capacity boundary" $x_1 + x_2 = 2$, which is $(0.95, 1.05)$, and the new optimal value is 3.05 instead of 3.
- But $3.05 = 3 + y_1^* \cdot (\text{change in available labor}) = 3 + 0.1y_1^*$.

To summarize, each dual variable represents the per unit increase in objective function value from relaxing one of the primal constraints for a *sufficiently small* amount, while other things held fixed.

- If each constraint of a linear program is interpreted as the restriction imposed by available quantity of some resource, the dual variable measures the *marginal value* of that resource.
- In other words, the dual variable is measuring the relative contribution that one unit of that resource makes to the total economic reward or revenue.

At the optimum, the two interpretation must coincide: the marginal cost must equal to the marginal revenue, as you learned in ECON1010 or ECON2010.

- For this reason, a solution to the dual, $\boldsymbol{y}^*$, is called the **shadow prices**.

Writing down the dual

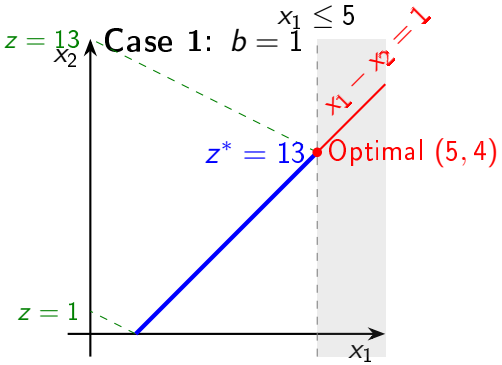
The interpretation of the dual allow us to answer the following questions regarding the dual.

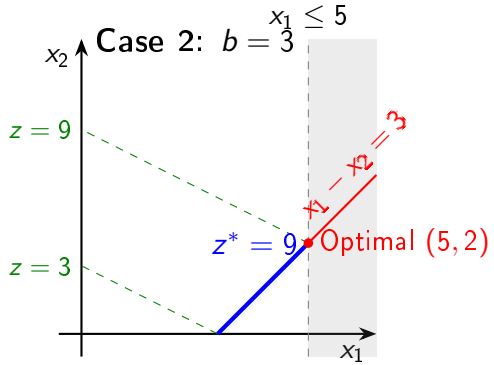
1. What happens to the dual if some constraints in the primal hold with $=$ instead of $\leq$ (we don't need to worry too much about $\geq$ because it can be easily converted to $\leq$)?

- Recall that the dual variable y_i for the constraint $\sum_{j=1}^{n} a_{ij}x_j \leq b_i$ must satisfy $y_i \geq 0$.
- This is because y_i can be interpreted as the marginal value of relaxing this constraint, which cannot be negative because doing so enlarges the constraint set.
- But if the constraint is $\sum_{j=1}^{n} a_{ij}x_j = b_i$, as b_i increases the constraint set becomes completely different, and therefore the marginal value can be negative.
- Thus, the sign of the dual variable for a equality constraint is *unconstrained*.

An implication is that the dual of the linear program $\max\{\boldsymbol{c} \cdot \boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \geq 0\}$ is

$$\min \boldsymbol{y} \cdot \boldsymbol{b} \qquad \text{s.t.} \quad \boldsymbol{y}\boldsymbol{A} \geq \boldsymbol{c}.$$





Maximize: $z = x_1 + 2x_2$
Subject to: $x_1 - x_2 = b$, $x_1, x_2 \geq 0$, $x_1 \leq 5$
Result: $b = 1 \Rightarrow z^* = 13$, but $b = 3 \Rightarrow z^* = 9$

2. What happens to the dual if some variables can also take negative values?
   - Recall that if we interpret the variable x_j as running activity j, $x_j \geq 0$ means that this activity must be run at a non-negative level.
   - In this case, the dual constraint for variable x_j: $\sum_{i=1}^{m} a_{ij} y_i \geq c_i$.
   - Inutuitively, this means that the marginal benefit from running activity j, c_j, the j-th entry of $\boldsymbol{c}$, must be no more than the imputed marginal cost of running this activity, $\sum_{i=1}^{m} a_{ij} y_i$.
   - Without the nonnegativity constraint, it can't be that $\sum_{i=1}^{m} a_{ij} y_i > c_i$: otherwise one would let $x_i \to -\infty$ to get infinite profits, which makes no sense.
   - Thus, we should expect the dual constraint for variable x_j to be $\sum_{i=1}^{m} a_{ij} y_i = c_i$.

An implication is that the dual of the linear program $\max\{\boldsymbol{c} \cdot \boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}\}$ is

$$\min \boldsymbol{y} \cdot \boldsymbol{b} \qquad \text{s.t.} \quad \boldsymbol{y}\boldsymbol{A} = \boldsymbol{c},\ \boldsymbol{y} \geq 0,$$

and the dual of the linear program $\max\{\boldsymbol{c} \cdot \boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}\}$ is

$$\min \boldsymbol{y} \cdot \boldsymbol{b} \qquad \text{s.t.} \quad \boldsymbol{y}\boldsymbol{A} = \boldsymbol{c}.$$

These can be derived using the tricks on slide 7.